FundBridge AI · System Architecture

Three-tier web application with an isolated AI/NLP service

PRESENTATION LAYER

React 19 single-page application · role-aware routing · responsive

- **Business Owner** — profile · readiness · matches
- **Investor** — discovery · fit · shortlist
- **Administrator** — verify · users · audit

HTTPS · httpOnly session cookie + CSRF token

APPLICATION LAYER

Node.js 24 · Express REST API · 64 endpoints

- **Middleware** — authenticate · role + ownership · CSRF · rate limit · validation
- **Services** — business · scheme · investor · proposal · RAG · audit · action plan
- **SCORING ENGINES** — readiness · scheme matching · investor matching · pure functions
- **Config service** — versioned weights · thresholds · vocabularies · reproducible

Prisma ORM

internal service token

DATA LAYER

PostgreSQL 18 · 21 tables · 21 enums

- User · BusinessProfile · InvestorProfile
- GovernmentScheme (verified, with provenance)
- FundingReadiness · SchemeMatch · InvestorMatch
- Proposal · ProposalAnalysis · ActionPlanItem
- SchemeDocumentChunk + 384-dim embeddings
- MatchingConfig (versioned) · AuditLog

AI / NLP SERVICE

Python 3.12 · FastAPI · isolated process

- Document extraction (PDF / DOCX)
- Text cleaning · section detection
- Sentence embeddings (all-MiniLM-L6-v2, CPU)
- Chunking + retrieval (cosine, thresholded)
- Proposal scoring
- LLM router · grounded generation only

optional

External LLM provider

phrases explanations only · never decides a score

Scores are produced only by the engines.

The platform stays fully usable when the AI service or the LLM is unavailable.